

Assignment 7 (Final Assignment)

February 9th, 2026

IMPORTANT INSTRUCTIONS

- This is the **final assignment** marking the end of our MCMC project.
- You must choose **one option only** from the three options below.
- **SUBMISSION FORMAT:**
 - Option 1: Submit one PDF file (LaTeX report compiled to PDF)
 - Option 2A or 2B: Submit one Python Notebook (.ipynb) file
 - **Important:** At the top of your .ipynb file, clearly mention which problem statement you are solving (2A or 2B)
 - Upload to Google Classroom
- For queries, contact us on WhatsApp.

CHOOSE ONE OPTION

Option 1: LaTeX Project Report

Write a complete LaTeX report documenting everything you learned in this MCMC project.

What to include:

- Briefly describe the objective behind this project.
- Briefly describe the theory you learnt (Markov Chains, MCMC, Ising Model etc.)
- Briefly mention about Your Ising model implementation and conclusion about phase transitions.
- Overall conclusion of the project.

Why this option: LaTeX is essential for scientific writing. This gives you hands-on experience with professional document preparation.

Submit: One PDF file (compiled from your .tex file). You can use any LaTeX editor (Overleaf, TeXstudio, etc.) and any template structure you prefer.

Option 2A: Traffic Jam Simulator (Physics)

Real-world context: Cities waste billions on traffic congestion. Better prediction models help optimize traffic lights and reduce jams.

Problem Statement:

Build a 1D highway simulation using Metropolis MCMC where cars maintain safe distances from neighbors. Start from one slow car and observe how traffic jams form and spread.

Hints - What to include in your notebook:

Note: These hints are suggestions to help you get started. You don't need to follow this structure rigidly. Feel free to implement the problem in any suitable way you choose, as this is an open-ended capstone task.

1. Create 1D array: 50 car positions (car #25 starts slow)
2. Energy function: penalize cars too close or too far from ideal spacing
3. Metropolis moves: try moving each car ± 1 position randomly

4. Plot 1: car positions vs time steps (visualize jam spreading)
5. Plot 2: highway snapshots at 5 different time points
6. Plot 3: jam size (cars within 3 spaces) vs time
7. Brief analysis: When does jam form? How does it grow?

Submit: One .ipynb file with your solution (mention “Problem 2A: Traffic Jam Simulator” at the top)

Option 2B: Crypto Portfolio Risk Calculator (Finance)

Real-world context: Cryptocurrency investors need accurate risk estimates for portfolio allocation. Simple calculations ignore uncertainty, leading to poor decisions.

Problem Statement:

Download real cryptocurrency price data for 5 coins. Use MCMC to estimate each coin’s daily return volatility from historical prices. Calculate portfolio risk for an equal-weight portfolio with uncertainty estimates.

Hints - What to include in your notebook:

Note: These hints are suggestions to help you get started. You don’t need to follow this structure rigidly. Feel free to implement the problem in any suitable way you choose, as this is an open-ended capstone task.

1. Download data: 30 days of BTC, ETH, ADA, SOL, LINK (you can use yfinance library)
2. Calculate daily log-returns from closing prices
3. MCMC target: volatility posterior for each coin from returns
4. Plot 1: MCMC trace for one coin’s volatility (show burn-in and sampling)
5. Table: estimated volatility mean and 95% confidence interval for all 5 coins
6. Final calculation: equal-weight portfolio volatility with uncertainty range
7. Brief analysis: Which coin has highest risk? How confident are you?

You can search these topics to get started: “Bayesian portfolio optimization MCMC”, “cryptocurrency volatility estimation”, “portfolio risk Python yfinance”, “MCMC finance tutorial”

Submit: One .ipynb file with your solution (mention “Problem 2B: Crypto Portfolio Risk” at the top)

KEY POINTS FOR OPTION 2 (CAPSTONE PROBLEMS)

Independent Learning Experience

No theory guides provided. You must:

- Search for relevant materials online (research papers, tutorials, GitHub repositories)
- Understand the problem domain on your own
- Implement MCMC creatively
- Debug and validate your results
- Ask doubts on WhatsApp if stuck

Open-Ended Nature

- The hints are **not mandatory steps** - they are suggestions only
- You can structure your solution however you prefer
- Different approaches are welcome
- Focus on solving the core problem using MCMC

Validation

- **Traffic:** Does your jam form realistically? Does it spread backward?
- **Crypto:** Do your volatility estimates make sense? Are confidence intervals reasonable?
- Include brief discussion validating your results

FINAL NOTES

This assignment tests your ability to learn independently. Either you will learn LaTeX or you will learn to apply MCMC to real problems. If you choose Option 2, you will discover that MCMC is a powerful tool used everywhere - from physics to finance.

Good luck with your learning experience!

For any doubts, contact us on WhatsApp.